

FINAL MARK

**GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION 2
HSC ASSESSMENT TASK 1, 2015 (HSC 2016)
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	E2	E3	E4	E5	E6	E7	E8	E9
Multiple choice	/5		✓						✓
Q6	/18		✓						✓
Q7	/24		✓						✓
Q8	/19		✓						✓
Q9	/20		✓						✓
Q10	/10		✓						✓
Q11	/11		✓						✓
TOTAL									
	/107		/107						/107

- E1 appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems.
- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.



GIRRAWEE HIGH SCHOOL

TASK 1

2015

MATHEMATICS

EXTENSION 2

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used.
- Start each question on a separate page. Each paper must show your name.

Multiple Choice (5 marks) Write the letter corresponding to the correct answer in your answer booklet.

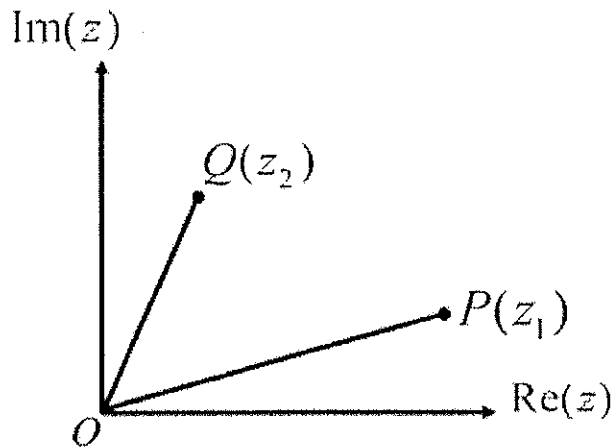
1. Find the conjugate of $\frac{1}{3+4i}$

- (A) $\frac{3}{25} - \frac{4}{25}i$ (B) $\frac{3}{25} + \frac{4}{25}i$ (C) $-\frac{3}{25} - \frac{4}{25}i$ (D) $-\frac{3}{25} + \frac{4}{25}i$

2. The cube roots of unity are 1, ω and ω^2 . Simplify: $\frac{1}{1+\omega} + \frac{1}{1+\omega^2}$

- (A) 0 (B) -1 (C) 1 (D) 2

3. The points P and Q in the first quadrant represent the complex numbers z_1 and z_2 respectively, as shown in the diagram below. Which statement about the complex number $z_2 - z_1$ is true?



- (A) It is represented by the vector QP .
- (B) Its principal argument lies between $\frac{\pi}{2}$ and π .
- (C) Its real part is positive
- (D) Its modulus is greater than $|z_1 + z_2|$.

4. Find $\arg(z^4)$ where $z = -4\sqrt{2} + 4\sqrt{2}i$

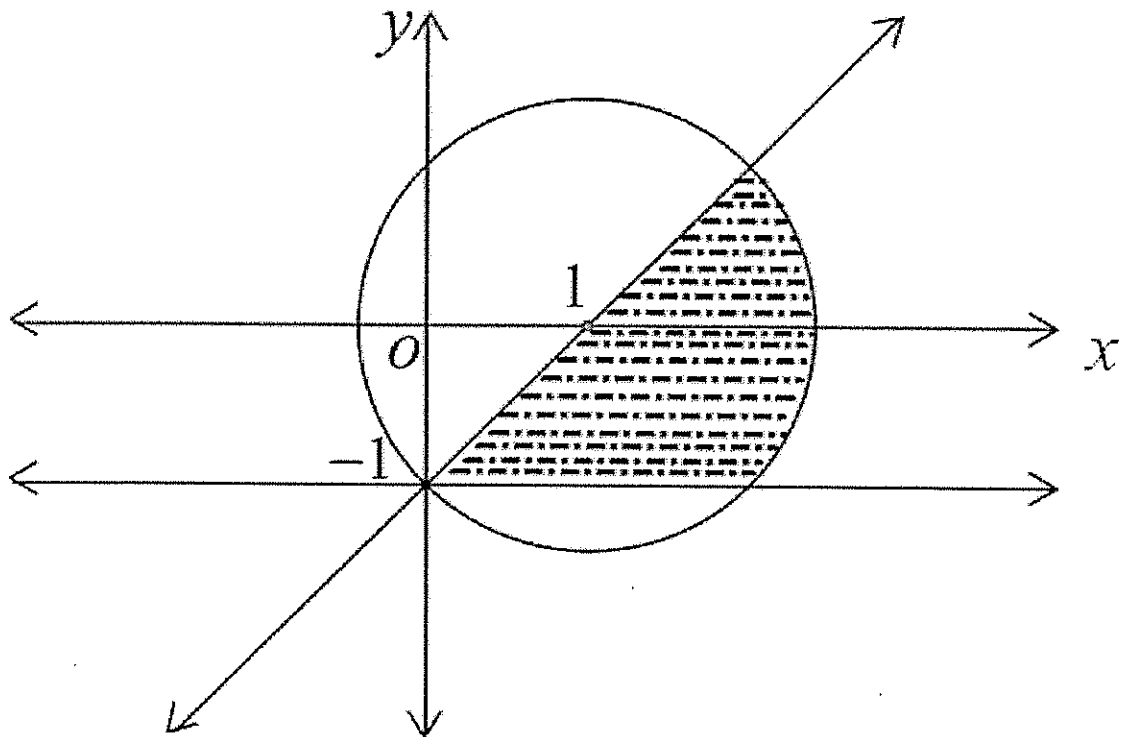
(A) π

(B) 3π

(C) $-\pi$

(D) $\frac{3\pi}{4}$

5. Consider the Argand diagram below. Which inequality could define the shaded area?



(A) $|z-1| \leq \sqrt{2}$ and $0 \leq \arg(z-i) \leq \frac{\pi}{4}$

(B) $|z-1| \leq \sqrt{2}$ and $0 \leq \arg(z+i) \leq \frac{\pi}{4}$

(C) $|z-1| \leq 1$ and $0 \leq \arg(z-i) \leq \frac{\pi}{4}$

(D) $|z-1| \leq 1$ and $0 \leq \arg(z+i) \leq \frac{\pi}{4}$

Question 6 (18 marks)

Marks

(a) If $z = (1-i)(2\sqrt{3} + 2i)$ (i) Express z in the form $x + iy$, where x and y are real. 2(ii) By expressing $1 - i$ and $2\sqrt{3} + 2i$ in the modulus argument form, show that

$$z = 4\sqrt{2}\text{cis}\left(-\frac{\pi}{12}\right)$$
3

(iii) Hence find the exact value of $\tan \frac{\pi}{12}$ 3(b) (i) Find all pairs of real numbers x and y such that $(x + iy)^2 = 8 + 6i$. 3(ii) Hence solve: $z^2 + 2(1 + 2i)z - (11 + 2i) = 0$ 3(c) Find the values of θ ($0 \leq \theta \leq 2\pi$) such that $\frac{3 + 2i \sin \theta}{1 - 2i \sin \theta}$ is a real number. 4**Question 7 (24 marks)**(a)(i) Express $\sin 6\theta$ and $\cos 6\theta$ in terms of $\cos \theta$ and $\sin \theta$. 4(ii) Hence express $\cot 6\theta$ in terms of $\cot \theta$. 3(b) (i) If $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$. 4(ii) Use De Moivre's theorem to obtain an expression for $\sin^7 \theta$ in the form $a \sin 7\theta + b \sin 5\theta + c \sin 3\theta + d \sin \theta$. What are the values of a, b, c and d . 4(c) (i) Solve $z^6 + 1 = 0$. Show the roots on an Argand diagram. 4(ii) Resolve $z^6 + 1$ into real quadratic factors. Hence show that

$$\cos 3\theta = 4 \cos \theta \left(\cos^2 \theta - \cos^2 \frac{\pi}{6} \right)$$
5

Question 8 (19 marks)

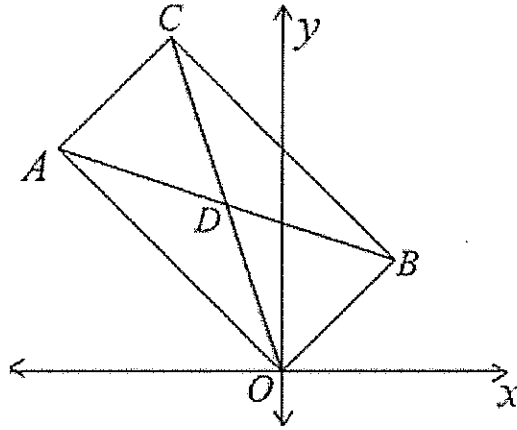
- (a) Find all the cube roots of i . Express your answer in the form $x + iy$. 3
- (b) (i) Find the roots of $z^7 = 1$. 2
- (ii) Show that the roots can be written in the form $1, \omega, \omega^2, \omega^3, \omega^4, \omega^5, \omega^6$ where $\omega = cis \frac{2\pi}{7}$. 2
- (iii) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$. 2
- (iv) Form a quadratic equation with roots $\omega + \omega^2 + \omega^4$ and $\omega^3 + \omega^5 + \omega^6$. 3
- (v) Hence find the exact value of $\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7}$. 4
- (c) Solve the equation $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$. 3

Question 9 (20 marks)

- (a) Sketch the following loci on separate Argand diagram.
- (i) $|z + 2 - i| = |z - 2 + i|$ 3
- (ii) $\arg(z - 1 - i) = \frac{\pi}{4}$ 3
- (iii) $1 \leq |z - 1| \leq 2$ 3
- (iv) $\arg(z + 2) - \arg(z - i) = \pi$ 3
- (b) Find the locus of z and sketch on an Argand diagram.
- (i) $|z - 2| = \operatorname{Re}(z)$ 4
- (ii) $\left| z^2 - \left(\frac{z}{z} \right)^2 \right| \geq 8$ 4

Question 10 (10 marks)

- (a) $OACB$ is a rectangle where $OA = 2 \times OB$. D is the point of intersection of the diagonals. The point B represents the complex number z .



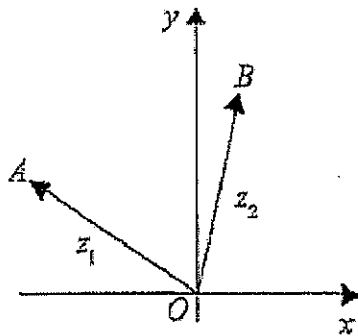
Find in terms of z , the complex number represented by:

(i) A

(ii) D

4

(b)



In the Argand diagram, vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers

$$z_1 = 2\left(\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}\right) \text{ and } z_2 = 2\left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15}\right) \text{ respectively.}$$

(i) Show that $\triangle OAB$ is equilateral.

3

(ii) Express $z_2 - z_1$ in modulus-argument form.

3

Question 11 (11 marks)

(a) Given the equation $\arg\left(\frac{z-3}{z+1}\right) = \frac{3\pi}{4}$.

(i) Draw $\overline{z-3}$, $\overline{z+3}$, $\arg(z-3)$ and $\arg(z+1)$ in an Argand diagram and mark the angle representing $\arg\left(\frac{z-3}{z+1}\right)$ giving reasons. 4

(ii) Find the equation of the locus of $\arg\left(\frac{z-3}{z+1}\right) = \frac{3\pi}{4}$, describe the locus and sketch in the Argand diagram drawn in (i). All reasons must be given. 7

END OF EXAMINATION

2015 Extension 2 Test 1 (2016 HSC)

1. Let $z = \frac{1}{3+4i}$

$$= \frac{1}{3+4i} \times \frac{3-4i}{3-4i}$$

$$= \frac{3-4i}{9-(4i)^2}$$

$$= \frac{3-4i}{9-16i^2}$$

$$= \frac{3-4i}{9+16}$$

$$= \frac{3-4i}{25}$$

$$= \frac{3}{25} - \frac{4i}{25}$$

$$\bar{z} = \frac{3}{25} + \frac{4}{25}i \quad \textcircled{B}$$

2. $\frac{1}{1+w} + \frac{1}{1+w^2}$

$$= \frac{1+w^2+1+w}{(1+w)(1+w^2)}$$

$$= \frac{1}{1+w^2+w+w^3}$$

$$= \frac{1}{0+1} = 1 \quad \textcircled{C}$$

3. B

4. Let $z = -4\sqrt{2} + 4\sqrt{2}i$

$$|z| = \sqrt{(4\sqrt{2})^2 + (4\sqrt{2})^2} = \sqrt{32+32} = 8$$

$$\tan \alpha = \frac{4\sqrt{2}}{4\sqrt{2}} = 1 \quad \alpha = \frac{\pi}{4}$$

$$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$-4\sqrt{2} + 4\sqrt{2}i = 8 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$z^4 = 8^4 \left(\cos 3\pi + i \sin 3\pi \right)$$

$$= 8^4 \left(\cos \pi + i \sin \pi \right) \quad \textcircled{A}$$

5. B

Question 6 (18 marks)

(a) i) $z = (1-i)(2\sqrt{3}+2i)$

$$= 2\sqrt{3} + 2i - i2\sqrt{3} - 2i^2 \quad \textcircled{2}$$

$$= 2\sqrt{3} + 2i - i2\sqrt{3} + 2$$

$$= 2 + 2\sqrt{3} + i(2 - 2\sqrt{3}) \quad \textcircled{1}$$

(ii) $z = 1-i$

$$|z| = \sqrt{1+1} = \sqrt{2}$$

$$\tan \alpha = 1$$

$$\alpha = \frac{\pi}{4}$$

$$\theta = -\frac{\pi}{4}$$

$$1-i = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)$$

$$w = 2\sqrt{3} + 2i$$

$$|w| = \sqrt{12+4} = 4$$

$$\tan \alpha = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}$$

$$2\sqrt{3} + 2i = 4 \operatorname{cis} \frac{\pi}{6}$$

$$(1-i)(2\sqrt{3}+2i)$$

$$= \left(\sqrt{2} \operatorname{cis} \frac{-\pi}{4} \right) \left(4 \operatorname{cis} \frac{\pi}{6} \right)$$

$$= 4\sqrt{2} \operatorname{cis} \left(\frac{\pi}{6} - \frac{\pi}{4} \right)$$

$$= 4\sqrt{2} \operatorname{cis} \left(\frac{2\pi - 3\pi}{12} \right)$$

(3)

$$= 4\sqrt{2} \operatorname{cis} \frac{-\pi}{12}$$

$$= 4\sqrt{2} \left(\cos \frac{\pi}{12} - i \sin \frac{\pi}{12} \right)$$

$$= 4\sqrt{2} \cos \frac{\pi}{12} - i 4\sqrt{2} \sin \frac{\pi}{12} \quad \text{--- (2)}$$

Equating real and imaginary parts of (1) and (2)

$$4\sqrt{2} \cos \frac{\pi}{12} = 2 + 2\sqrt{3}$$

$$\cos \frac{\pi}{12} = \frac{2 + 2\sqrt{3}}{4\sqrt{2}}$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$$-4\sqrt{2} \sin \frac{\pi}{12} = 2 - 2\sqrt{3}$$

$$\sin \frac{\pi}{12} = \frac{2 - 2\sqrt{3}}{-4\sqrt{2}}$$

$$= \frac{2\sqrt{3} - 2}{4\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\tan \frac{\pi}{12} = \frac{\sin \frac{\pi}{12}}{\cos \frac{\pi}{12}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \times \frac{2\sqrt{2}}{1 + \sqrt{3}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

(3)

$$(x+iy)^2 = 8+6i$$

$$x^2 + 2ixy - y^2 = 8+6i$$

$$x^2 - y^2 = 8$$

$$2xy = 6$$

$$xy = 3$$

$$y = \frac{3}{x}$$

$$x^2 - \frac{9}{x^2} = 8$$

$$x^4 - 9 = 8x^2$$

$$x^4 - 8x^2 - 9 = 0$$

$$\begin{matrix} pq = -9 \\ p+q = -8 \end{matrix}$$

$$(-9, 1)$$

$$m^2 - 8m - 9 = 0 \quad (3)$$

$$(m-9)(m+1) = 0$$

$$m = 9 \text{ or } m = -1$$

$$x^2 = 9 \text{ or } x^2 = -1$$

$$x = \pm 3 \quad (\because x \text{ is real})$$

$$\text{When } x = 3, y = 1$$

$$\text{When } x = -3, y = -1$$

$\therefore$ the square roots are

$$3+i \text{ or } -3-i$$

$$\pm (3+i)$$

page 3

(ii)

$$Z = \frac{-2(1+2i) \pm \sqrt{4(1+2i)^2 - 4 \times -(11+2i)}}{2}$$

$$= \frac{-2(1+2i) \pm \sqrt{4(1+4i+4i^2) + 4(11+2i)}}{2}$$

$$= \frac{-2-4i \pm \sqrt{4+16i-16+44+8i}}{2}$$

$$= \frac{-2-4i \pm \sqrt{32+24i}}{2}$$

$$= \frac{-2-4i \pm \sqrt{4(8+6i)}}{2}$$

$$= \frac{-2-4i \pm 2\sqrt{8+6i}}{2} \quad (3)$$

$$= \frac{-2-4i \pm 2(3+i)}{2}$$

$$= \frac{-2-4i+2(3+i)}{2} \text{ or } \frac{-2-4i-2(3+i)}{2}$$

$$= \frac{-2-4i+6+2i}{2} \text{ or } \frac{-2-4i-6-2i}{2}$$

$$= \frac{4-2i}{2} \text{ or } \frac{-8-6i}{2}$$

$$= \frac{2(2-i)}{2} \text{ or } \frac{2(-4-3i)}{2}$$

$$= 2-i \text{ or } -4-3i$$

$$(c) \frac{3+2i\sin\theta}{1-2i\sin\theta}$$

$$= \frac{(3+2i\sin\theta)(1+2i\sin\theta)}{(1-2i\sin\theta)(1+2i\sin\theta)}$$

$$= \frac{(3+2i\sin\theta)(1+2i\sin\theta)}{1-(2i\sin\theta)^2}$$

$$= \frac{3+6i\sin\theta+2i\sin\theta+4i^2\sin^2\theta}{1-4i^2\sin^2\theta}$$

$$= \frac{3+8i\sin\theta-4\sin^2\theta}{1+4\sin^2\theta}$$

$$= \frac{3-4\sin^2\theta+8i\sin\theta}{1+4\sin^2\theta}$$

$$= \frac{3-4\sin^2\theta}{1+4\sin^2\theta} + \frac{8\sin\theta}{1+4\sin^2\theta} i$$

Since $\frac{3+2i\sin\theta}{1-2i\sin\theta}$ is purely real, its imaginary part is 0

$$\frac{8\sin\theta}{1+4\sin^2\theta} = 0$$

$$8\sin\theta = 0$$

$$\sin\theta = 0$$

$$\theta = \underline{\underline{0, \pi, 2\pi}}$$

(4)

Question 7 (23 marks)

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$$(1) (\cos \theta + i \sin \theta)^6 = \cos 6\theta + i \sin 6\theta \quad \text{by De Moivre's theorem} \quad \textcircled{1}$$

Apply Binomial theorem,

$$(\cos \theta + i \sin \theta)^6 = \cos^6 \theta + 6C_1 \cos^5 \theta i \sin \theta + 6C_2 \cos^4 \theta i^2 \sin^2 \theta + 6C_3 \cos^3 \theta i^3 \sin^3 \theta + 6C_4 \cos^2 \theta i^4 \sin^4 \theta + 6C_5 \cos \theta i^5 \sin^5 \theta + 6C_6 i^6 \sin^6 \theta$$

$$= \cos^6 \theta + 6 \cos^5 \theta i \sin \theta + 15 \cos^4 \theta \times -1 \sin^2 \theta - i 20 \cos^3 \theta \sin^3 \theta + 15 \cos^2 \theta \sin^4 \theta + 6 \cos \theta \times i \sin^5 \theta + -1 \times \sin^6 \theta \quad \textcircled{2}$$

Equating real and imaginary parts of ① and ②

$$\cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$$

$$\sin 6\theta = 6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta$$

$$\cot 6\theta = \frac{\cos 6\theta}{\sin 6\theta}$$

$$= \frac{\cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta}{6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta}$$

divide both numerator and denominator by $\sin^6 \theta$

$$= \frac{\cot^6 \theta - 15 \cot^4 \theta + 15 \cot^2 \theta - 1}{6 \cot^5 \theta - 20 \cot^3 \theta + 6 \cot \theta} \quad \textcircled{3}$$

$$(b)(i) \quad z = \cos \theta + i \sin \theta$$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos n\theta + i \sin n\theta \quad \text{by De Moivre's theorem}$$

$$\frac{1}{z^n} = z^{-n} = (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos(-n\theta) + i \sin(-n\theta) \quad \text{by De Moivre's}$$

$$= \cos n\theta - i \sin n\theta \quad \left(\because \sin(-\alpha) = -\sin \alpha \right. \\ \left. \cos(-\theta) = \cos \theta \right)$$

$$z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$= \underline{\underline{2 \cos n\theta}}$$

(4)

$$z^n - \frac{1}{z^n} = \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta$$

$$= \underline{\underline{2i \sin n\theta}}$$

$$(ii) \quad (2i \sin \theta)^7 = \left(z - \frac{1}{z} \right)^7$$

$$= z^7 - {}^7C_1 z^6 \times \frac{1}{z} + {}^7C_2 z^5 \times \frac{1}{z^2} - {}^7C_3 z^4 \times \frac{1}{z^3}$$

$$+ {}^7C_4 z^3 \times \frac{1}{z^4} - {}^7C_5 z^2 \times \frac{1}{z^5} + {}^7C_6 z \times \frac{1}{z^6} - {}^7C_7 \frac{1}{z^7}$$

$$= z^7 - 7z^5 + 21z^3 - 35z + 35 \frac{1}{z} - 21 \times \frac{1}{z^3} + 7 \times \frac{1}{z^5} - \frac{1}{z^7}$$

$$= \left(z^7 - \frac{1}{z^7} \right) - 7 \left(z^5 - \frac{1}{z^5} \right) + 21 \left(z^3 - \frac{1}{z^3} \right) - 35 \left(z - \frac{1}{z} \right)$$

$$= 2i \sin 7\theta - 7 \times 2i \sin 5\theta + 21 \times 2i \sin 3\theta - 35 \times 2i \sin \theta$$

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$$2^7 i^7 \sin^7 \theta = 2i \sin 7\theta - 14i \sin 5\theta + 42i \sin 3\theta - 70i \sin \theta$$

$$2^7 x - i \sin^7 \theta = 2i \sin 7\theta - 14i \sin 5\theta + 42i \sin 3\theta - 70i \sin \theta$$

$$2^6 x - i \sin^7 \theta = \sin 7\theta - 7 \sin 5\theta + 21 \sin 3\theta - 35 \sin \theta$$

$$-64 \sin^7 \theta = \sin 7\theta - 7 \sin 5\theta + 21 \sin 3\theta - 35 \sin \theta$$

$$\sin^7 \theta = \frac{-1}{64} [\sin 7\theta - 7 \sin 5\theta + 21 \sin 3\theta - 35 \sin \theta]$$

$$= \frac{1}{64} [-\sin 7\theta + 7 \sin 5\theta - 21 \sin 3\theta + 35 \sin \theta]$$

$$= \frac{-1}{64} \sin 7\theta + \frac{7}{64} \sin 5\theta - \frac{21}{64} \sin 3\theta + \frac{35}{64} \sin \theta$$

$$a = \frac{-1}{64} \quad b = \frac{7}{64} \quad c = -\frac{21}{64} \quad d = \frac{35}{64} \quad (4)$$

$$(c) z^6 + 1 = 0$$

$$z^6 = -1 = \cos \pi + i \sin \pi$$

$$= \cos(\pi + 2k\pi) + i \sin(\pi + 2k\pi), k \in \mathbb{I}$$

$$= \cos((2k+1)\pi) + i \sin((2k+1)\pi), k \in \mathbb{I}$$

$$z = [\cos((2k+1)\pi) + i \sin((2k+1)\pi)]^{\frac{1}{6}}$$

$$= \cos \frac{(2k+1)\pi}{6} + i \sin \frac{(2k+1)\pi}{6}, k = 0, 1, 2, 3, 4, 5$$

(by De Moivre's theorem)

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When $k=0$,

$$z_1 = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

When $k=1$,

$$z_2 = \cos \frac{3\pi}{6} + i \sin \frac{3\pi}{6}$$

$$= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \quad (\text{K})$$

When $k=2$,

$$z_3 = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}$$

When $k=3$,

$$z_4 = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}$$

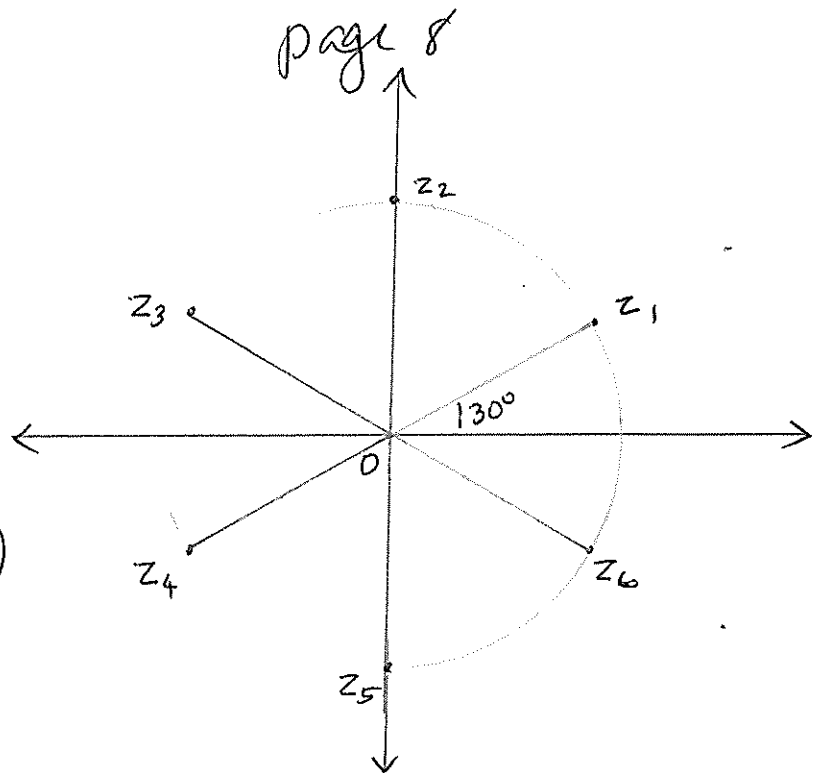
When $k=4$,

$$z_5 = \cos \frac{9\pi}{6} + i \sin \frac{9\pi}{6}$$

$$= \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} = -i$$

When $k=5$,

$$z_6 = \cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}$$



From the diagram

$$z_6 = \overline{z_1}, z_5 = \overline{z_2}, z_4 = \overline{z_3}$$

$$z_1 + z_6 = z_1 + \overline{z_1} = 2 \cos \frac{\pi}{6}$$

$$z_1 z_6 = z_1 \overline{z_1} = 1$$

$$z_3 + z_4 = z_3 + \overline{z_3} = 2 \cos \frac{5\pi}{6}$$

$$z_3 z_4 = z_3 \overline{z_3} = 1$$

$$\begin{aligned}
z^6 + 1 &= (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)(z - z_6) \\
&= (z - z_1)(z - z_6)(z - z_3)(z - z_4)(z - z_2)(z - z_5) \\
&= (z - z_1)(z - z_6)(z - z_3)(z - z_4)(z - i)(z + i) \\
&= (z^2 + 1)(z^2 - z(z_1 + z_6) + z_1 z_6)(z^2 - z(z_3 + z_4) + z_3 z_4) \\
&= (z^2 + 1)(z^2 - z \times 2 \cos \frac{\pi}{6} + 1)(z^2 - z \times 2 \cos \frac{5\pi}{6} + 1) \\
z^6 + 1 &= (z^2 + 1)(z^2 - 2 \cos \frac{\pi}{6} z + 1)(z^2 - 2 \cos \frac{5\pi}{6} z + 1)
\end{aligned}$$

divide both sides by z^3

$$\begin{aligned}
\frac{z^6 + 1}{z^3} &= \frac{z^2 + 1}{z} \times \frac{z^2 - 2 \cos \frac{\pi}{6} z + 1}{z} \times \frac{z^2 - 2 \cos \frac{5\pi}{6} z + 1}{z} \\
z^3 + \frac{1}{z^3} &= \left(z + \frac{1}{z}\right) \left(z - 2 \cos \frac{\pi}{6} + \frac{1}{z}\right) \left(z - 2 \cos \frac{5\pi}{6} + \frac{1}{z}\right) \\
2 \cos 3\theta &= 2 \cos \theta \left(2 \cos \theta - 2 \cos \frac{\pi}{6}\right) \left(2 \cos \theta - 2 \cos \frac{5\pi}{6}\right) \\
2 \cos 3\theta &= 8 \cos \theta \left(\cos \theta - \cos \frac{\pi}{6}\right) \left(\cos \theta - \cos \frac{5\pi}{6}\right) \\
\cos 3\theta &= 4 \cos \theta \left(\cos \theta - \cos \frac{\pi}{6}\right) \left(\cos \theta - \cos \frac{5\pi}{6}\right) \\
&= 4 \left(\cos \theta - \cos \frac{\pi}{2}\right) \left(\cos \theta - \cos \frac{\pi}{6}\right) \left(\cos \theta - \cos \frac{5\pi}{6}\right) \\
&= 4 \left(\cos \theta - \cos \frac{\pi}{2}\right) \left(\cos \theta - \cos \frac{\pi}{6}\right) \left(\cos \theta + \cos \frac{\pi}{6}\right) \\
&= 4 \left(\cos \theta - \cos \frac{\pi}{2}\right) \left(\cos^2 \theta - \cos^2 \frac{\pi}{6}\right) \quad \text{⑤} \quad \begin{aligned} \therefore \cos \frac{5\pi}{6} &= -\cos \left(\pi - \frac{5\pi}{6}\right) \\ &= -\cos \frac{\pi}{6} \end{aligned}
\end{aligned}$$

Question 8 (19 marks)

$$\begin{aligned}
 (a) \quad z^3 = i &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \\
 &= \cos \left(2k\pi + \frac{\pi}{2} \right) + i \sin \left(2k\pi + \frac{\pi}{2} \right) \\
 &= \text{Cis } \frac{4k\pi + \pi}{2} \\
 &= \text{Cis } \frac{(4k+1)\pi}{2}
 \end{aligned}$$

$$z = \text{Cis } \frac{(4k+1)\pi}{6} \quad k=0,1,2 \quad \text{by De Moivre's Theorem}$$

when $k=0$,

$$\begin{aligned}
 z_1 &= \text{Cis } \frac{\pi}{6} \\
 &= \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \\
 &= \frac{\sqrt{3}}{2} + i \times \frac{1}{2} = \frac{\sqrt{3}}{2} + \frac{1}{2}i
 \end{aligned}$$

when $k=1$

$$\begin{aligned}
 z_2 &= \text{Cis } \frac{5\pi}{6} \\
 &= \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \\
 &= -\frac{\sqrt{3}}{2} + i \times \frac{1}{2} \\
 &= -\frac{\sqrt{3}}{2} + \frac{1}{2}i
 \end{aligned}$$

when $k=2$

$$\begin{aligned}
 z_3 &= \text{Cis } \frac{9\pi}{6} = \text{Cis } \frac{3\pi}{2} \\
 &= -i
 \end{aligned}$$

The cube roots are

$$\underline{\underline{-i, \pm \frac{\sqrt{3}}{2} + \frac{i}{2}}}$$

(3)

(b) (i) $z^7 = 1 = \text{cis } 0 = \text{cis } 2k\pi, k \in \mathbb{Z}$ page 11

$$z = [\text{cis } 2k\pi]^{\frac{1}{7}}$$

$$= \text{cis } \frac{2k\pi}{7} \quad k=0, 1, 2, 3, 4, 5, 6, \text{ by De Moivre's theorem.}$$

Roots are

$$1, \text{cis } \frac{2\pi}{7}, \text{cis } \frac{4\pi}{7}, \text{cis } \frac{6\pi}{7}, \text{cis } \frac{8\pi}{7}, \text{cis } \frac{10\pi}{7}, \text{cis } \frac{12\pi}{7}$$

(ii) let $w = \text{cis } \frac{2\pi}{7}$

$$w^2 = \left(\text{cis } \frac{2\pi}{7}\right)^2 = \text{cis } \frac{4\pi}{7}$$

$$w^3 = \left(\text{cis } \frac{2\pi}{7}\right)^3 = \text{cis } \frac{6\pi}{7}$$

$$w^4 = \left(\text{cis } \frac{2\pi}{7}\right)^4 = \text{cis } \frac{8\pi}{7}$$

$$w^5 = \left(\text{cis } \frac{2\pi}{7}\right)^5 = \text{cis } \frac{10\pi}{7}$$

$$w^6 = \left(\text{cis } \frac{2\pi}{7}\right)^6 = \text{cis } \frac{12\pi}{7}$$

$\therefore$ the roots can be written as $1, w, w^2, w^3, w^4, w^5$ and w^6 .

(iii) $1 + w + w^2 + w^3 + w^4 + w^5 + w^6$

$$= -\frac{\text{Coefficient of } z^6}{\text{Coefficient of } z^7} = -\frac{0}{1} = 0$$

(iv) $z_1 = w + w^2 + w^4, \quad z_2 = w^3 + w^5 + w^6$

$$z_1 + z_2 = w + w^2 + w^4 + w^3 + w^5 + w^6 = -1$$

$$Z_1 Z_2 = (\omega + \omega^2 + \omega^4)(\omega^3 + \omega^5 + \omega^6)$$

$$= \omega^4 + \omega^6 + \omega^7 + \omega^5 + \omega^7 + \omega^8 + \omega^7 + \omega^9 + \omega^{10}$$

$$= \omega^4 + \omega^6 + 1 + \omega^5 + 1 + \omega + 1 + \omega^2 + \omega^3$$

$$= 0 + 2 = 2$$

The quadratic equation with roots Z_1 and Z_2 is

$$Z^2 - (-1)Z + 2 = 0$$

$$Z^2 + Z + 2 = 0$$

(3)

$$(V) Z^2 + Z + 2 = 0$$

$$Z = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times 2}}{2}$$

$$= \frac{-1 \pm \sqrt{1-8}}{2} = \frac{-1 \pm \sqrt{-7}}{2}$$

$$= \frac{-1 \pm i\sqrt{7}}{2}$$

$$= \frac{-1}{2} + \frac{i\sqrt{7}}{2} \text{ or } \frac{-1}{2} - \frac{i\sqrt{7}}{2} \quad \text{--- (1)}$$

$$\text{Consider } Z_1 = \omega + \omega^2 + \omega^4$$

$$= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$$

$$= \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} \right) + i \left(\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} \right) \quad \text{--- (2)}$$

Comparing (1) and (2) we get

$$\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} = \frac{\sqrt{7}}{2} \text{ or } -\frac{\sqrt{7}}{2} \quad \text{--- (3)}$$

$$\text{But } \sin \frac{4\pi}{7} = \sin\left(\pi - \frac{4\pi}{7}\right) = \sin \frac{3\pi}{7}$$

$$\sin \frac{8\pi}{7} = -\sin\left(\frac{8\pi}{7} - \pi\right) = -\sin \frac{\pi}{7}$$

(3) becomes

(4)

$$\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2} \text{ or } -\frac{\sqrt{7}}{2}$$

$$\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} > 0$$

$$\therefore \sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$

$$(C) \quad z^5 + z^4 + z^3 + z^2 + z + 1 = 0$$

$$z^6 - 1 = (z - 1)(z^5 + z^4 + z^3 + z^2 + z + 1)$$

$$z^5 + z^4 + z^3 + z^2 + z + 1 = \frac{z^6 - 1}{z - 1}$$

$$z^6 - 1 = 0, z \neq 1$$

$$z^6 = 1 = \cos 0 + i \sin 0$$

$$= \cos 2k\pi + i \sin 2k\pi \quad k \in \mathbb{Z}$$

$$z = \left(\cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6} \right) \quad k = 0, 1, 2, 3, 4, 5$$

by De Moivre's theorem

$$= \cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3} \quad k = 0, 1, 2, 3, 4, 5$$

when $k=0$, $z_1 = 1$

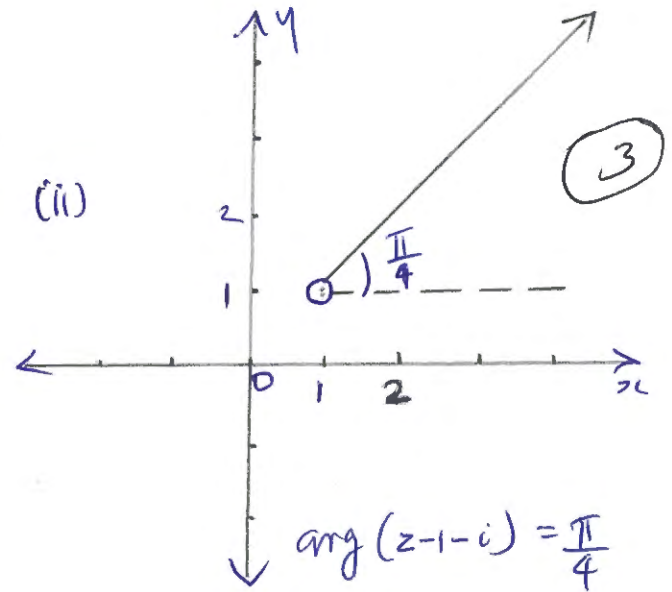
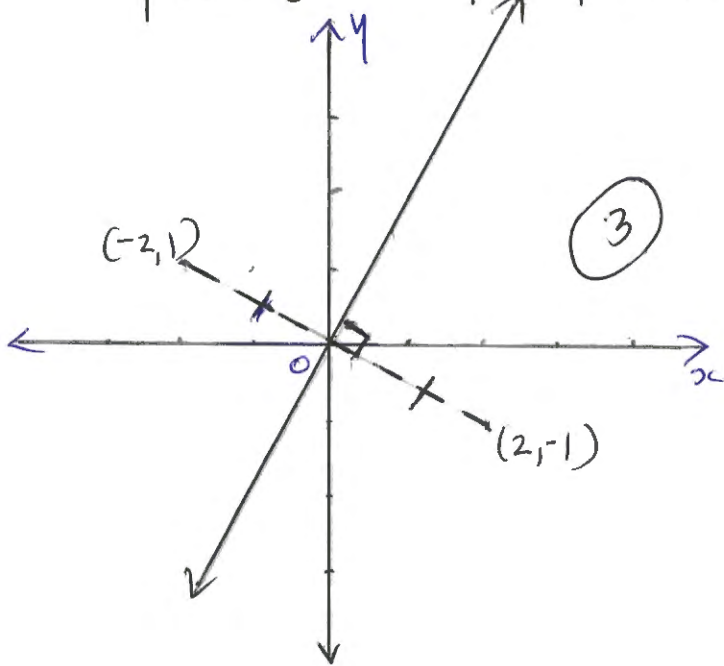
$\therefore$ the roots of $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ are

given by $z = \cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3}, \quad k = 1, 2, 3, 4, 5$

Question 9 (20 marks)

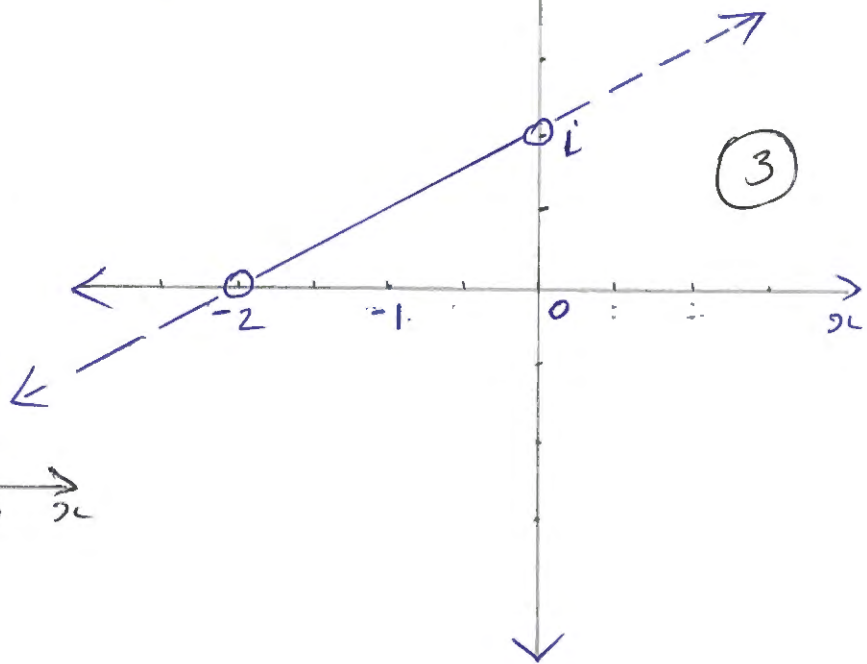
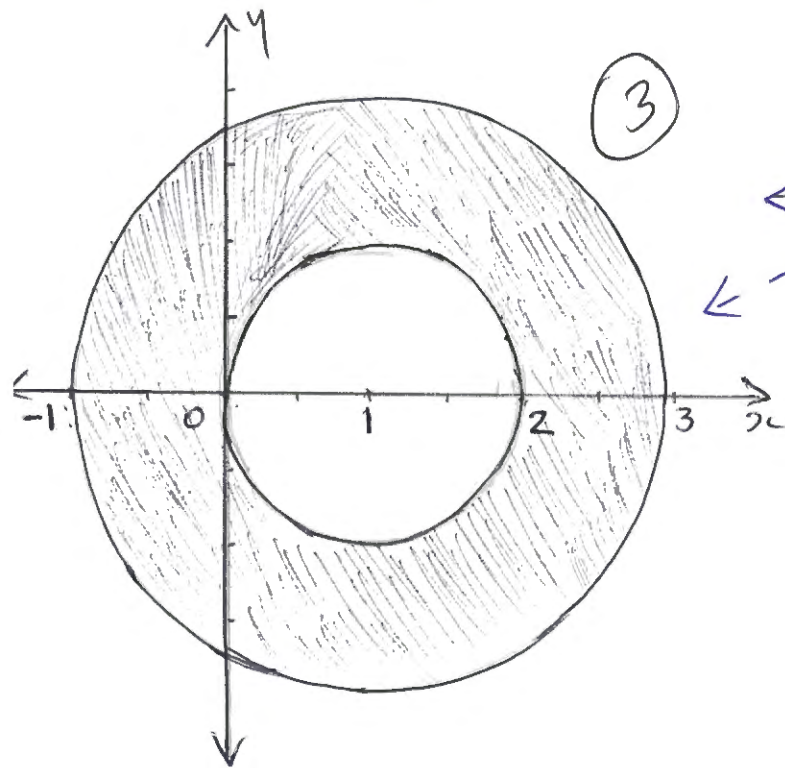
(a)(i) $|z+2-i| = |z-2+i|$

$$|z - (-2+i)| = |z - (2-i)|$$



(iv)

(iii) $1 \leq |z-1| \leq 2$



$$\arg(z+2) - \arg(z-i) = \pi$$

(b) $|z - z| = \operatorname{Re}(z)$

$$|x + iy - 2| = x$$

$$|x - 2 + iy| = x$$

$$\sqrt{(x-2)^2 + y^2} = x$$

$$(x-2)^2 + y^2 = x^2 \quad (4)$$

$$x^2 - 4x + 4 + y^2 = x^2$$

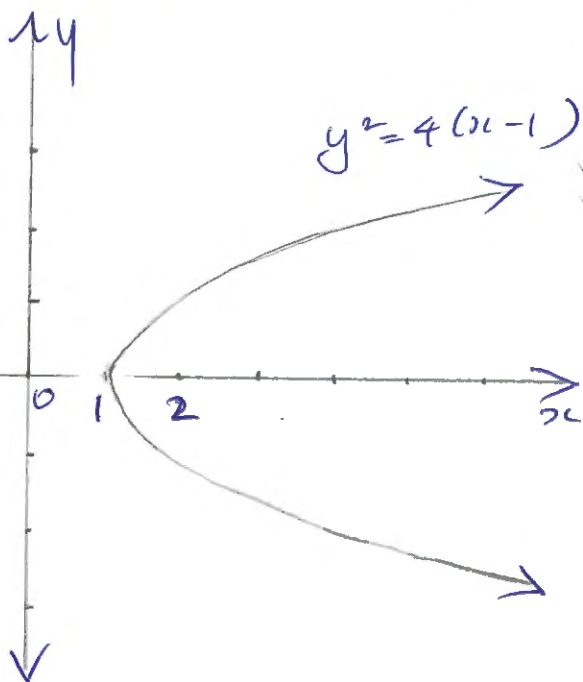
$$y^2 = 4x - 4$$

$$= 4(x-1)$$

$$4a = 4 \quad a = 1$$

Vertex (1,0) Focus (2,0)

Directrix $x = 0$



(ii) $|z^2 - \bar{z}^2| \geq 8$

$$|(x+iy)^2 - (x-iy)^2| \geq 8$$

$$|(x+iy+x-iy)(x+iy-x+iy)| \geq 8$$

$$|(2x)(2iy)| \geq 8$$

$$|4ixy| \geq 8$$

$$|4i| |xy| \geq 8$$

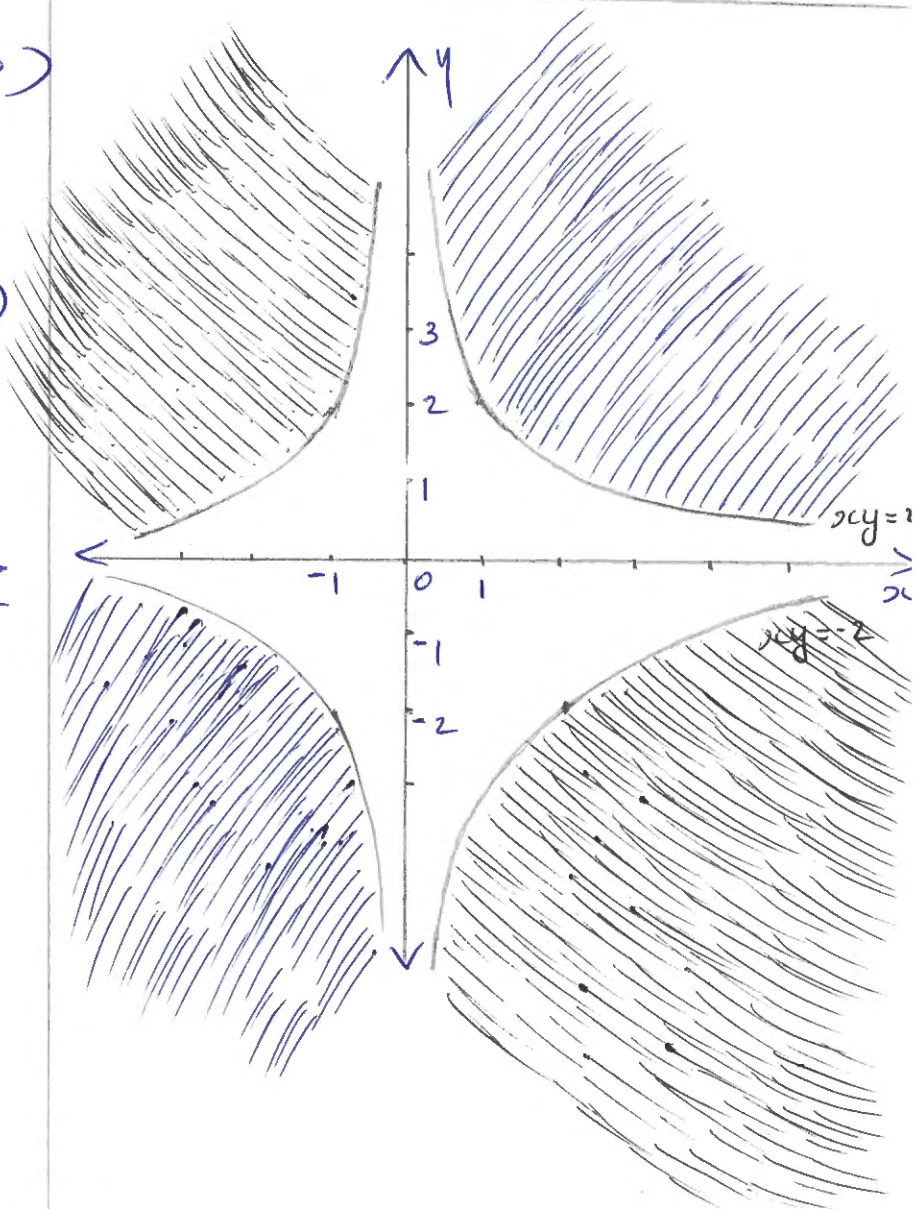
$$4 |xy| \geq 8$$

$$|xy| \geq 2$$

$$xy \geq 2 \text{ or } -xy \geq 2$$

$$xy \geq 2 \text{ or } xy \leq -2$$

(4)



Question 10 (10 marks)

page 16

(a) (i) $OA = 2 \times OB$

$$\vec{OA} = 2z$$

$\therefore A$ is $2z$

$$\vec{OC} = \vec{OA} + \vec{AC}$$

$$= 2z + z$$

$$= (1+2i)z$$

(1)

(ii) D is the midpoint of

O and C ($\because$ diagonals bisect each other)

$$D = \frac{(1+2i)z}{2}$$

$$= \frac{1}{2}z + iz$$

(3)

(b) (i) $\angle AOB = \angle AOX - \angle BOX$

$$= \arg(z_1) - \arg(z_2)$$

$$= \frac{4\pi}{5} - \frac{7\pi}{15} = \frac{\pi}{3}$$

$$|z_1| = |z_2| = 2$$

(3)

$$\angle OAB = \angle OBA = \frac{\pi - \frac{\pi}{3}}{2} = \frac{\pi}{3}$$

(angles opposite equal sides in an isosceles $\triangle$)

$\therefore \triangle OAB$ is equilateral.

(ii) $z_2 - z_1$ is represented by $\vec{AB}$.

AB is the rotation of z_2 in the clockwise direction through an angle of $\frac{\pi}{3}$

(3)

$$\vec{AB} = OB \times \text{cis}\left(-\frac{\pi}{3}\right)$$

$$z_2 - z_1 = z_2 \text{cis}\left(-\frac{\pi}{3}\right)$$

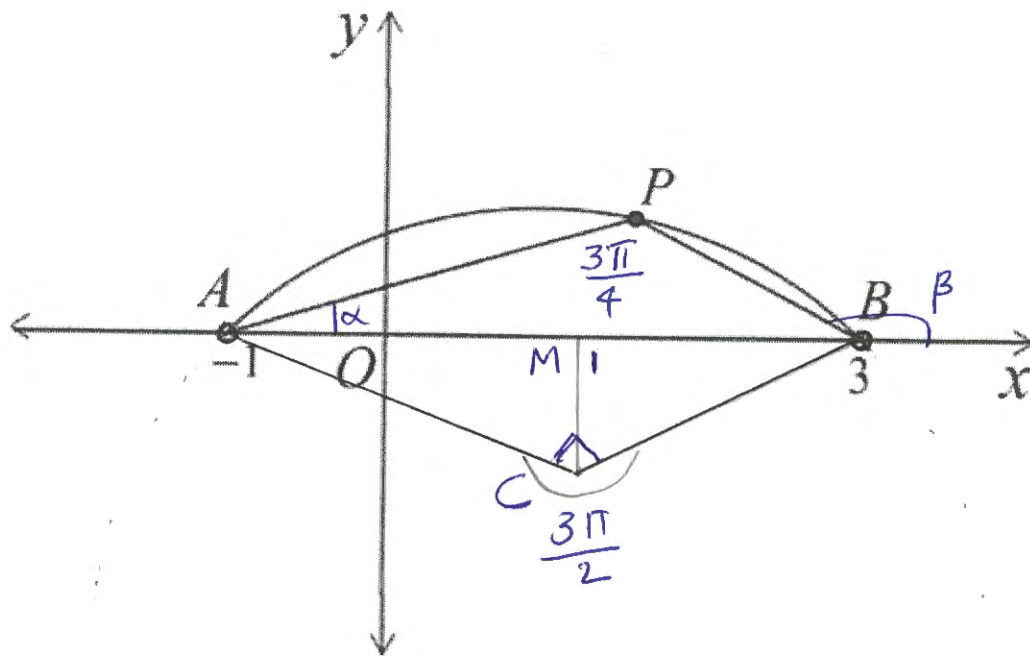
$$= 2 \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right)$$

$$\left(\cos -\frac{\pi}{3} + i \sin -\frac{\pi}{3} \right)$$

$$= 2 \text{cis}\left(\frac{7\pi}{15} - \frac{\pi}{3}\right)$$

$$= 2 \text{cis}\left(\frac{2\pi}{15}\right)$$

$$= 2 \left(\cos \frac{2\pi}{15} + i \sin \frac{2\pi}{15} \right)$$



$$(i) \arg(z-3) = \angle PBX = \beta$$

$$\arg(z+1) = \angle PAX = \alpha$$

$$\text{Let } \angle APB = \theta$$

$$\alpha + \theta = \beta \text{ (exterior angle of } \triangle APB \text{)}$$

$$\theta = \beta - \alpha$$

$$= \arg(z-3) - \arg(z+1)$$

$$= \arg\left(\frac{z-3}{z+1}\right) = \frac{3\pi}{4}$$

(4)

(ii) Let C be the centre of the circle. Draw

$$CM \perp AB$$

$M = \text{Midpoint of } AB = (1, 0)$

(The perpendicular from the centre of a circle to a chord bisects the chord)

Reflex $\angle ACB = \frac{3\pi}{2}$ (angle at the centre is twice angle at the circumference)

$\triangle ACB$ is isosceles ($AC = BC = \text{radii}$)

$$\begin{aligned}\angle CBA &= \frac{\pi - \frac{\pi}{2}}{2} \quad (\text{angles opposite equal sides in an isosceles } \triangle) \\ &= \frac{\pi}{4}\end{aligned}$$

In $\triangle CBM$

$$\tan \frac{\pi}{4} = \frac{CM}{BM} = \frac{CM}{2}$$

$$\frac{CM}{2} = 1$$

$$CM = 2$$

centre $(1, -2)$

Apply Pythagoras' theorem

in $\triangle CBM$

$$BC = \sqrt{2^2 + 2^2}$$

$$= \sqrt{8}$$

$$\text{radius} = \sqrt{8}$$

locus is the minor arc of the circle

$$\underline{\underline{(x-1)^2 + (y+2)^2 = 8, y > 0}}$$

⑦